

Vikram Anantha

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Education

University of Michigan

Aug. 2024 – May 2028 (expected)

B.S. Computer Science & Electrical Engineering

3.94/4.00 GPA

- **Relevant coursework:** Logic Design, Differential Equations, Machine Learning, Computer Vision, Data Structures & Algorithms, Digital Circuits, Computer Organization, Linear Algebra, Discrete Math
- **Clubs:** Head Photo Editor of the Michiganensian Yearbook, Active member of Tau Beta Pi Engineering Honors Society
- **Honors:** 1 of 6 recipients of 1931e Merit Scholarship for Engineering, 3-term James B. Angell Scholar, Tau Beta Pi Scholarship

Lexington High School

Sept. 2020 – Jun. 2024

Lexington, MA

4.00/4.00 GPA

- 36/36 ACT score • 5/5 on AP Computer Science A, AP Calculus BC, AP Physics C, AP Chem

Industry Experience

Meta

May. 2026 – Jul. 2026

Software Eng. Intern

Menlo Park, CA

- Created a novel prototype of Server Side Rendering for React Native mobile development on Meta's family of apps
- Reduced bundle size by 30%, reached out to teams to measure impact and collect feedback on prototype

Capoom

Jan. 2025 – Dec. 2025

Software Eng. Intern, via Perot Jain Tech Lab at U of M's Center for Entrepreneurship

Ann Arbor, MI

- Architected Gaussian Splatting pipeline for 3D models of UofM's Mcity, cutting AV training data costs
- Built hierarchical Gaussian Splat map visualization, managing a team of 3 student engineers

Stellantis

May 2025 – Aug. 2025

ADAS Functional Integration Eng. Intern

Auburn Hills, MI

- Deployed CAPL script to analyze CAN-bus signals, identifying a 10% fault rate for ADAS teams
- Designed MATLAB Simulink model to control CAN-bus signals, quantifying self-driving behavior

Constant Therapy Health

Jun. 2024 – Aug. 2024

Software Eng. Intern

Newton, MA

- Fine-tuned Wav2Vec ASR on aphasia phonemes, achieving 4x lower false negative rate than Google's
- Built task visualizer and LLM grammar detector to support stroke patient speech therapy

HELM Learning

Apr. 2020 – May 2024

Co-Founder, Instructor, Developer

Lexington, MA

- Founded peer-to-peer learning platform with 4,200+ students and 90+ teachers across 10+ countries
- Built full-stack platform on Flask, AWS, and MySQL with MLP recommendation for 90+ classes

KTBYTE

Apr. 2018 – May 2024

Instructor

Lexington, MA

- Lead instructor & curriculum developer of Python, Java, Arduino classes for 100+ students, ages 8-18
- Robotics class lead instructor, mentored students to create inventions w/ Arduino + Raspberry Pi; 15+ students won awards nationally

Quantum-Enhanced ML Algorithm to Detect Alzheimer's

Oct. 2021 – May 2022

Project Creator

- Developed Quantum Algorithm to diagnose Alzheimer's Disease in patient with >85% accuracy
- Presented and won at MSEF Regionals and States 2022

My Automated Communication Companion

Feb. 2019 – Sept. 2019

Project Creator

- Built smart-hat using object detection and speech recognition to help non-verbal autistic people learn to speak
- Won awards at Invention Convention Worldwide, Young Inventors Program, and MSEF States 2019

Research Experience

University of Michigan, ROAHM Robotics Lab	Aug. 2024 – Present
- Built real-time 3D segmentation (Gaussian Splat, SAM-2, CLIP) via ROS2 for safe robotic arm use - Developing Transformer-based VLA models for generalist robot policies across diverse tasks	
Columbia University, Center for Smart Streetscapes	May 2023 – Feb. 2024
- Developed CV + MQTT app to warn pedestrians of collisions; co-authored 2024 IEEE IV Symposium paper - Optimized alert pipeline to <1.5s latency, meeting human reaction time to prevent accidents	
Massachusetts Institute of Technology, Auto ID Lab	Jun. 2023 – Sep. 2023
Built YOLOv8-based 2FA system to verify AV headlight patterns, securing V2I communication	
Massachusetts Institute of Technology, MetaConscious Group	Jul. 2022 – Aug. 2022
Created a Reinforcement Learning agent to mimic primates when they learn under a changing environment	

Photography Experience

Michiganensian Yearbook	Sept. 2024 – Present
<i>Head Photo Editor</i>	<i>University of Michigan, Ann Arbor, MI</i>
Manage photographic workflow, organize assignments, deadlines, credentials; lead meetings for development and critiques, proof and re-edit for aesthetic coherence	
Lexington Observer	Jun. 2023 – Present
<i>Staff Photographer</i>	<i>Lexington, MA</i>
Official Photographer for the Town of Lexington MA's online journal; 50+ photos; My articles: lexobserver.org/author/vikram-anantha	
Town of Lexington, Lexington Public Schools	Jun. 2023 – Present
<i>Freelance Photographer</i>	<i>Lexington, MA</i>
Official Photographer for the Lexington MA's town government and public schools; 100+ photos	
Lexington High School Yearbook + newspaper <i>The Musket</i>	Sept. 2022 – Jun. 2024
<i>Head Photo Editor</i>	<i>Lexington High School, Lexington, MA</i>
Manage photographic workflow, organize assignments, deadlines, credentials; lead meetings for development and critiques, proof and re-edit for aesthetic coherence	

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, CAPL, ReactJS, Assembly, Bash, LaTeX
Frameworks & Tools: Robot Operating System (ROS & ROS2), Amazon Web Services (AWS), Vector CANalyzer, MATLAB/Simulink, OpenCV, TensorFlow/PyTorch, Git/GitHub, CUDA, Docker, Gaussian Splatting, LLMs, YOLOv8 Object Detection, SAM-2 Segmentation, SLAM

Selected Honors & Awards

Immerse The Bay Hackathon, Stanford University, 1st Place Winner	Nov. 2025
Lexington Public Schools Core Values award, Receipt, awarded by LPS superintendent	Oct. 2023
Diamond Challenge Business Competition, 1st Place Winner for HELM Learning	Mar. 2023
Drexel University High School Photo Contest, Selected for Photo Exhibition	Dec. 2022 & 2023
Kurt Giessler Foundation, Prize for Achievement for HELM Learning (\$750 grant money)	Oct. 2022
Mass. Science & Engineering Fair (MSEF), Outstanding AI Project Winner	Mar. & May 2022
American Invitational Mathematics Exam (AIME), 2 Time Qualifier	Mar. 2021 & Feb. 2022
Presidential Volunteer Service Award (PVSA), Gold Award Recipient	Nov. 2021
USA Coding Olympiad (USACO), Silver Division	Dec. 2019
Invention Convention Worldwide (ICW), UTC Best in Show: Most Innovative Project	May 2019

Publications

Digital Twin for Pedestrian Safety Warning at a Single Urban Traffic Intersection	2024
V. Anantha, et al., 2024 IEEE Intelligent Vehicles Symposium (IV), pp. 2640-2645, doi: 10.1109/IV55156.2024.10588544	
Peer to Peer Learning Platform Optimized With Machine Learning	2022
V. Anantha, arXiv:2209.03489, 2022	